Cheatsheet AZ-305 — 02 · Load balancers and Virtual WAN

1. The four load balancers

	Azure Load Balancer	Application Gateway	Azure Front Door	Traffic Manager
Scope	Regional	Regional	**Global**	**Global**
OSI layer	4 (TCP/UDP)	7 (HTTP/HTTPS)	7 (HTTP/HTTPS)	DNS (does not see traffic)
Protocol	Any TCP/UDP	HTTP(S) only, WebSocket	HTTP(S) only	**Any** (agnostic)
Sees the traffic	Yes (packets)	Yes (proxy)	Yes (proxy at the edge)	**No** (only resolves DNS)
SSL termination / offloading	✗	✓	✓ (at the edge)	✗
URL / host-based routing	✗	✓	✓	✗
WAF (OWASP)	✗	✓ (WAF v2)	✓	✗
mTLS (client cert)	✗	✓	✗	✗
Session affinity	Client IP	Cookie	Cookie	✗
Health probe	TCP / HTTP	HTTP(S) per path	HTTP(S) layer 7	Basic HTTP / TCP
Private IP / internal	✓ (Internal LB)	✓ (private frontend)	✗ (always public)	✗
Backends	VMs, VMSS, AKS nodes	VMs, VMSS, App Service, AKS (AGIC), IPs	App Service, AKS, VMs, Storage, other LBs	Any endpoint with IP/DNS
Typical case	SQL AG listener (ILB + Floating IP), L4 balancing	Regional web app with WAF and SSL	Global multi-region app with centralized SSL	Global DNS failover, non-HTTP protocols

Traffic Manager methods: Priority (active/standby) · Weighted · Performance (latency) · Geographic · MultiValue · Subnet. **Traffic Manager front-end failover** → **Endpoint monitor settings** (interval, timeout, tolerated failures).

2. How to choose in two questions

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1. Is it HTTP(S)?
   ├─ No (SQL, RDP, FTP, generic TCP)
   │    ├─ Regional → Load Balancer
   │    └─ Global → Traffic Manager (the only global non-HTTP option)
   └─ Yes → 2.
2. One region or several?
   ├─ One region → Application Gateway (L7, WAF, SSL, mTLS, private IP)
   └─ Several regions → Front Door (L7, SSL at the edge, WAF, CDN)
                        Traffic Manager if DNS is enough and centralized SSL is not needed
```

Valid combinations: Front Door → App Gateway (regional) → VMs. Traffic Manager → App Gateway. Front Door → public Load Balancer.

3. Application Gateway: exam details

- **End-to-end SSL:** terminates TLS, **re-encrypts** towards the backend (backend HTTP setting with certificate) — vs **SSL offloading**: terminates and sends in clear text.
- **WAF v2:** OWASP CRS + custom rules incl. **rate limiting** (recent).
- **AGIC:** App Gateway as AKS ingress controller.
- Needs its **own (dedicated) subnet**.
- **v2** required for autoscaling, zone redundancy, private frontend.

4. Front Door: exam details

- SSL termination at global **points of presence**; managed certificates; custom domains.
- Automatic failover via **L7 health probes**; minimum latency via anycast.
- Integrated cache / CDN; WAF; rate limiting.
- Has **no** private IP; protects the backend with X-Azure-FDID / Private Link to origin (Premium).
- Decisive signal: **"SSL without configuring it on each backend"** + multi-region.

5. Load Balancer: exam details

- **Standard** SKU: zone-redundant, HA ports, NSG required, SLA 99.99 %. **Basic**: retired.
- **Internal LB for SQL AG listener:** enable **Floating IP** (DSR); **TCP** probe on the listener port; session persistence **does not apply**.
- **Outbound rules / NAT Gateway** for Internet egress.
- Pair of NVAs in HA → behind an **Internal LB** (avoids the UDR single point of failure).

6. Virtual WAN

Capability	Basic	Standard
Site-to-Site VPN	✓	✓
ExpressRoute	✗	✓
Point-to-Site VPN	✗	✓
Hub-to-hub (multi-region)	✗	✓
VNet ↔ VNet transit via hub	✗	✓
Secured virtual hub (Azure Firewall in the hub)	✗	✓
Third-party NVA in the hub	✗	✓
Upgrade Basic → Standard	Yes (mandatory first step before ExpressRoute)	—

- **Hubs:** one per region where presence is required ("supporting connection in three regions" → 3 hubs). Circuits and offices are spread across the existing hubs.
- **Global VPN profile (P2S):** the client connects **automatically to the nearest hub**.
- **Transitive routing:** native in Virtual WAN Standard; **classic peering is not transitive**.
- **FQDN filtering:** only **Azure Firewall** (secured hub) — NSGs and ASGs filter by IP/port.

7. Signals → answer

Wording	Answer
"regional", "path-based routing", "WAF", "SSL offloading"	Application Gateway
"multi-region" + "SSL managed centrally"	Front Door
"multi-region" + "any protocol" / "DNS"	Traffic Manager
"control the threshold for failing over the front end"	Traffic Manager endpoint monitor
"layer 4", "TCP/UDP", "internal", "SQL listener"	Load Balancer (+ Floating IP)
"single private IP + mTLS + rate limiting" (APIM options)	APIM Premium Internal (exam answer)
"ExpressRoute" + "Virtual WAN"	Standard
"P2S closest region", "transitive", "FQDN filtering"	Virtual WAN Standard + secured hub
"connection in N regions" → hubs	N hubs

8. Mistakes to avoid

1. Choosing Traffic Manager when **centralized SSL** is required → it does not see the traffic.
2. Choosing Load Balancer to route by URL → it is layer 4.
3. Choosing Front Door for **private** access → it is always public.
4. HTTP probe on the SQL listener → TCP.
5. Session persistence on the ILB of an AG listener → what is missing is Floating IP.
6. Virtual WAN Basic with ExpressRoute → impossible; Standard.
7. Counting circuits or offices as hubs → count the **required regions**.